

Key: P price · Q quantity · MB, MC marginal benefit, cost · MR marginal revenue · MU marginal utility · OC opportunity cost · PPC production possibilities curve · CS, PS consumer, producer surplus · DWL deadweight loss · S, P, E social, private, external · ϵ price elasticity · % Δ percent change · F25 = the Fall 2025 midterm · "Problem Set" is always written out, because PS means producer surplus

How he grades

- 60 minutes for 60 points with a hard stop, so do the easy parts first. Algebra is used, calculus is not.
- Write inside the answer box in 2 to 6 sentences. His rule: "Less is more."
- "Explain" means give the reason, not just the number. Use the words his key rewards: marginal benefit, marginal cost, willingness to pay, opportunity cost.
- Each graph is worth 4 points. Put price on the vertical axis, label every curve, and mark the equilibrium at the question's numbers with dotted lines to both axes.
- If you get stuck, write your reasoning in words for partial credit. If a question is unclear, say how you read it.

Inelastic means steep and elastic means flat.

Opportunity cost and the PPC

- Opportunity cost (OC) is the value of the best alternative given up.
- Your time, even free time, is worth about your wage. At the average wage of \$36 an hour, 3 hours cost about \$108, so a \$200 concert ticket is expensive.
- The OC of using a ticket you own is its resale value today plus your time. The price you paid is sunk, so ignore it.
- Example: an afternoon makes 2 bookcases or 3 TV shows, so one show costs 2/3 of a bookcase.

The OC of college is tuition, fees, books, and foregone earnings (about \$60,000 a year). Leave out rent and food, since you pay them whether or not you go.

- TRAP** Mistake: listing only the costs you pay. Correct: full credit requires the words "foregone earnings," which exceed tuition except at private colleges.

PPC slope = minus the OC of the good on the horizontal axis. Flatter means that good has a lower OC.

- A point on the PPC is efficient. A point inside means idle resources. A point outside cannot be reached.
- TRAP** Mistake: a recession (like COVID) shifts the PPC in. Correct: it moves the economy to a point inside the PPC.
- Bowed out means OC rises as you make more, because people differ in skill. Straight means constant OC.
- More workers, more capital, or better technology shift the PPC out. Investing today shifts it out later.

Comparative advantage

Whoever has the lower OC of a good has the comparative advantage in it and should make it.

- TRAP** Mistake: picking whoever makes more per hour (absolute advantage). Correct: compare OCs. "The task I do best" means my lowest-OC task.

Per hour	Fish	Coconuts	1 coconut costs	1 fish costs
Bill	1	1	1 fish	1 coconut
Chris	8	2	4 fish	1/4 coconut
Robinson	2	2	1 fish	1 coconut

- Example (F25): Bill gathers coconuts and Chris fishes. Robinson ties Bill at 1 fish per coconut, so both gather (Problem Set 1).
- Example (F25): a fishhook doubles everyone's fish per hour, so a coconut now costs Bill 2 fish and Chris 8. Bill's OC is still lower, so nobody switches.

10 hours each	Max fish, coconuts	Straight slope	Specialized: kink (coconuts, fish)
Bill, Chris	90, 30	-3	-1, kink (10, 80), -4
All three	110, 50	-2.2	-1, kink (30, 80), -4

Both PPCs share the intercepts. The specialized PPC bows out above the straight one (nobody specializing), because the lowest-OC gatherers switch first.

- Example: at 10 coconuts, Bill and Chris get 80 fish when they specialize and 60 when they do not.
- Example: Tom makes 24 nuts or 12 coffee a day, Krishna 12 nuts or 24 coffee. Alone, each gets 8 of each. Specialized and trading 1 for 1, each gets 12 of each. Joint PPC: 36 coffee, kink at (24 nuts, 24 coffee), 36 nuts.
- Trade helps only when OCs differ. Both sides gain at any price between their two OCs (Bill and Chris: between 1 and 4 fish per coconut).
- Exception: a builder should work on his own house, because a hired crew has less reason to be careful.
- TRAP** Mistake: choosing who does a task by who can afford it. Correct: choose by OC.

Supply and demand

Demand slopes down because marginal benefit (willingness to pay) falls with each extra unit: "more is better, but at a decreasing rate."

- Market demand adds up every buyer's quantity at each price. Add sideways, never add prices.
- Supply slopes up because MC rises: extra output comes from producers with a higher opportunity cost.

A change in the good's own price moves you along the curves. Only other causes shift a curve.

- Below equilibrium, the shortage (quantity demanded minus supplied, $Q_D - Q_S$) bids the price up. Above it, the surplus ($Q_S - Q_D$) pushes it down.

Shift	P	Q	Caused by
Demand right	up	up	richer buyers, pricier substitute, cheaper complement, in fashion
Demand left	down	down	new or cheaper substitute, pricier complement, out of fashion
Supply right	down	up	better technology, cheaper inputs, more sellers
Supply left	up	down	pricier inputs, a ban, sellers leave, disease (bird flu kills hens)

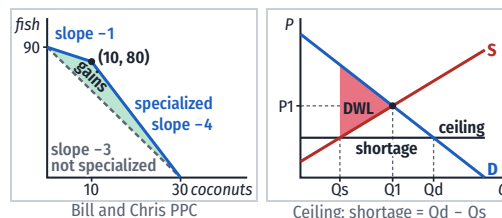
- TRAP** Mistake: reading "supply shifts up" as more supply. Correct: up means left, because MC is higher at every Q. Down means right.
- If both curves shift the same way, you know which way Q moves but not P. If they shift opposite ways, you know which way P moves but not Q.
- Reading data: if price and quantity both rose (or both fell), demand shifted. If they moved in opposite directions, supply shifted.
- A reason about buyers moves only demand. A reason about sellers moves only supply. Example: "fewer people need gas" shifts demand, not supply.
- When one curve shifts, the flatter the other curve, the more Q changes and the less P changes. If supply is vertical (a crop already planted), only P changes.
- Example (Problem Set 1): new light rail makes commuting cheaper, so housing demand rises. Houston's loose building rules make supply flat, so mostly more homes get built. In California supply is steep, so mostly rents rise.

Surplus and efficiency

- CS = area below demand and above the price: MB minus price on each unit.
- PS = area above supply and below the price. Sellers get the price on each unit but pay only its MC, so PS = revenue - variable cost.**
- If supply is flat and price = MC, then PS = 0.
- Total surplus = CS + PS, the area between the curves up to the quantity traded. With a tax, add the tax revenue. With a subsidy, subtract its cost.
- Triangle = $\frac{1}{2} \times \text{base} \times \text{height}$. Trapezoid = average of the two parallel sides $\times$ width.
- Efficient means total surplus is as large as possible. That needs Q where MB = MC, with the highest-MB buyers buying and the lowest-MC sellers selling.
- Given lists of buyer values and seller costs: pair the highest values with the lowest costs while value $\geq$ cost. The number of pairs is the efficient Q.

DWL = surplus lost when Q is above or below the efficient Q. The DWL triangle points toward the efficient Q.

- First Welfare Theorem: a competitive market with no externalities maximizes total surplus. It ignores fairness, because willingness to pay depends on income.



Price controls

- A ceiling (legal maximum) only matters below equilibrium. A floor (legal minimum) only matters above it.
- TRAP** Mistake: shifting a curve for a rent ceiling. Correct: neither curve shifts. The law only caps the price.
- Binding ceiling:** Q traded = Q_S , and $Q_S < Q_1$ (old equilibrium) $< Q_D$. Shortage = $Q_D - Q_S$. DWL = triangle from Q_S to Q_1 .
- Total surplus always falls. CS can rise or fall: buyers gain a rectangle from PS (lower price on units still sold) but lose a triangle (units no longer sold).**
- Since lines or rationing, not price, decide who buys, some units go to lower-MB buyers. That is a loss on top of the DWL.
- Rent control helps tenants who get a unit and shields them from rent spikes. It hurts newcomers and landlords. Alternative: public housing.
- Binding floor: Q traded = Q_D , and the surplus is $Q_S - Q_D$. DWL is at least the triangle, more if higher-MC sellers get the sales.
- Anti-gouging laws (Problem Set 1): the model predicts a shortage, and high-MB buyers may miss out. It leaves out that buyers value fairness and that sellers may have market power (hand sanitizer early in the pandemic).

Elasticity

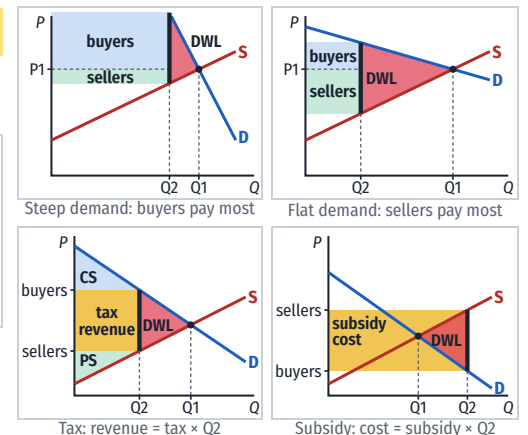
Elasticity is how much quantity responds to price: $\epsilon = \% \Delta Q \div \% \Delta P$, with each % change taken from the starting value.

- From a line: $\epsilon = (1 \div \text{slope}) \times (P \div Q)$. Demand $P = a - bQ$ gives $\epsilon = -(1 \div b) \times P \div Q$.
- Demand ϵ is negative. Keep the sign, but judge by size: -0.5 has size 0.5, under 1, so inelastic.
- A straight demand line is more elastic at higher prices. Example: $P = 12 - Q$ gives $\epsilon = -2$ at $Q = 4$, -1 at $Q = 6$, and -0.5 at $Q = 8$.

Spending = $P \times Q$, so $\% \Delta \text{ spending} \approx \% \Delta P + \% \Delta Q$.

Size of demand ϵ	Curve	If price rises 1%, spending
0	vertical	rises 1%
below 1	inelastic, steep	rises ($\epsilon = -0.75 \div +0.25\%$)
1	unit elastic	does not change
above 1	elastic, flat	falls ($\epsilon = -1.5 \div -0.5\%$)
infinite	horizontal	falls to zero

- To find a price change, flip it: $\% \Delta P = \% \Delta Q \div \epsilon$. Example: a supply shock cuts Q by 10% and demand $\epsilon = -0.5$, so price rises 20%.
- Example (F25): lunch price \$10 to \$11 is +10%, quantity 50,000 to 47,500 is -5%. $\epsilon = -5\% \div 10\% = -0.5$, inelastic.
- Demand is inelastic when buyers have no close substitute, the good is a need, or time is short (gas, eggs).
- Demand is elastic with close substitutes or more time. A narrow good is more elastic than its category: one lunch spot is about -1.5, but eating lunch out at all is -0.5.
- Supply is inelastic in the short run (crops already planted) and more elastic in the long run.
- TRAP** Mistake: extra spending = price rise $\times$ old quantity. Correct: Q falls when price rises, so this overstates spending unless demand is vertical.
- Example: for a big price change, compute $P \times Q$ before and after. Gas \$3 to \$4 (+33%) with $\epsilon = -0.37$: Q falls 12%, from 50 to 44 gallons, so spending goes from \$150 to about \$175.



Taxes and subsidies

- A per-unit tax is a vertical gap equal to the tax at the new quantity Q_2 (Q_1 is the no-tax quantity). Buyers pay P_D , read off demand. Sellers keep $P_S = P_D - \text{tax}$.
- TRAP** Mistake: who legally pays decides who bears the tax. Correct: a tax on sellers shifts supply up by the tax, a tax on buyers shifts demand down by it, and both give the same prices.

The more inelastic side (less able to walk away) bears more of the tax. Similar elasticities split it about evenly.

- Buyers bear all of it if supply is flat or demand is vertical. Sellers bear all of it if supply is vertical or demand is flat. A vertical curve means Q does not change, so $DWL = 0$.

Tax	Why the curve has that shape	Bears most
Berkeley soda	Coke and Pepsi can sell in nearby cities instead, so supply to Berkeley is flat	buyers (all)
Local beer	Brews are planned in advance, so supply is steep	brewery
Payroll	The law splits it evenly, but labor supply is steep (people work at almost any wage)	workers
EU gas	Stations pay the world oil price, so supply is flat	drivers

- TRAP** Mistake: answering "both pay" on the soda tax with no reason. Correct: it earns credit only if you argue supply is not flat (for example, a statewide tax).

Revenue = tax $\times$ Q_2 . $DWL = \frac{1}{2} \times \text{tax} \times (Q_1 - Q_2)$.

- Example: demand $P = 10 - Q$, supply $P = Q$, so $Q_1 = 5$. \$2 tax: set demand price = supply price + tax, $10 - Q = Q + 2$, so $Q_2 = 4$. Buyers pay 6, sellers keep 4, revenue 8, $DWL = 1$. \$2 subsidy: $10 - Q = Q - 2$, so $Q_2 = 6$, buyers pay 4, sellers get 6, cost 12, $DWL = 1$.
- A tax causes no misallocation: the units still sold go to the buyers who value them most. A price ceiling can misallocate.
- DWL is larger when curves are more elastic, because Q falls more. So taxing inelastic goods (cigarettes) is more efficient.
- A subsidy is a negative tax. Cost = subsidy $\times$ Q_2 . $DWL = \frac{1}{2} \times \text{subsidy} \times (Q_2 - Q_1)$, because the extra units cost more than buyers value them ($MC > MB$).
- In Finland, a sales tax cut on haircuts lowered prices by only about half the cut, but a later increase passed through almost fully.
- Scatter of price against tax across countries: slope 1 means buyers pay all the tax, 0 means sellers do, and above 1 means price rose by more than the tax. EU gas (Problem Set 1) is about 1.3, because richer countries have higher costs and higher taxes.

Households

- Budget line, with food (f) on the vertical axis and clothing (c) on the horizontal: $P_f \times q_f + P_c \times q_c = \text{income}$. Slope = $-P_c \div P_f$.
- Example: income \$100, food \$20, clothing \$50. The line runs from 5 food to 2 clothing, with slope -2.5 .
- If income rises, the line shifts out in parallel. If clothing's price rises, the line pivots in around the food intercept and the consumer is worse off.
- Both prices doubling with income fixed is the same as halving income.

Rational Spending Rule: $MU_a / P_a = MU_b / P_b$. The last dollar spent on each good buys the same MU.

- TRAP** Mistake: "good a got pricier, so MU_a falls." Correct: you buy less of a , so MU_a rises until the rule holds again. If a gets cheaper, buy more of a , so MU_a falls.
- Example with food at \$10 ($MU = 2$) and clothing at \$50 ($MU = 10$): each gives 0.2 MU per dollar. If clothing rises to \$55, $10 \div 55 = 0.18$, so buy less clothing until its MU is 11.
- Example (Problem Set 2): a heatwave raises electricity's MU at every quantity, so electricity demand shifts out. Buy more electricity and less of everything else.
- MU is the extra utility from one more unit. It falls as you consume more. The height of the demand curve = MU = willingness to pay.
- If MU falls fast (food), quantity barely responds to income or price. If it falls slowly (clothing), quantity responds a lot.
- Example (Problem Set 2): with utility = $\ln(c)$ (the $\ln$ key), MU = $1/c$. Alice with \$400 has MU 0.0025, Bob with \$160 has 0.00625. Their combined utility peaks when their MUs are equal: \$280 each, a \$120 transfer.

Competitive firms

- Profit = revenue – total cost, and cost includes opportunity costs such as the owner's own wage. Zero profit means the owner earns just what she could earn elsewhere.
- A competitive firm is a price taker: it is too small to move the market price, so its own demand is flat and $MR = P$.
- Produce where $MR = MC$. For a price taker that means $P = MC$, so its supply curve is its MC curve.**
- Fixed costs (rent, the farm itself) stay the same at any output, so they do not change MC. MC is the extra variable cost of one more unit.
- Industry supply adds all firms' MC curves sideways. Example: 100 firms each making 1,000 at \$5 supply 100,000 at \$5.
- A firm enters only if its PS (revenue – variable cost, the area between P and MC) beats its fixed cost.
- TRAP** Mistake: calling PS profit. Correct: profit = PS – fixed cost.
- If an input gets pricier, every firm's MC rises and market supply shifts left. P rises less than MC did, so each firm makes less and some exit. Cheaper technology does the reverse.
- Long run: profit attracts new firms and losses drive firms out until profit is zero, unless barriers block entry.

Monopoly

- A firm has market power when it can raise its price without losing all its customers. That happens when buyers have no close substitute.
- Example (F25): there is only one Taylor Swift, so fans have no substitute and she can charge far above MC.
- Barriers to entry keep rivals out: big fixed costs (railways), patents (drugs), licenses. A natural monopoly (like PG&E's power lines) can serve the whole market more cheaply than several firms.

Step 1: find Q where $MR = MC$. Step 2: go straight up from that Q to the demand curve and read the price.

- TRAP** Mistake: reading the price where MR crosses MC. Correct: the price comes from demand. In the class notebook example, $MR = MC = 2$ at $Q = 8$, so $P = 4$, not 2.
- MR is below price because every buyer pays the same price. To sell one more unit the firm must cut the price on all units, so it loses revenue on the earlier ones.

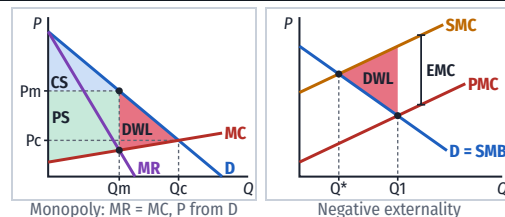
MR starts at the same intercept as demand and is twice as steep: $P = a - bQ$ gives $MR = a - 2bQ$, which hits zero halfway to demand's Q intercept.

- With constant $MC = c$: monopoly $Q_m = (a - c) \div 2b$ and $P_m = (a + c) \div 2$. The efficient Q , where demand meets MC, is twice Q_m .
- Monopoly sells less and charges more than a competitive market. DWL is the triangle between demand and MC from Q_m out to the efficient Q .
- Part of the competitive CS becomes profit: the rectangle between the competitive price and P_m , out to Q_m . That is a transfer from buyers to the firm, not a loss.
- Class notebook: demand $P = 6 - Q/4$, $MR = 6 - Q/2$, $MC = Q/4$. Swift (F25 Q3), from the key's graph: $P = 300 - 10Q$, $MR = 300 - 20Q$, $MC = 10Q$, with Q in millions and CS, PS, DWL in \$ billions.

Case	Q	P	CS	PS	DWL
Notebook, competitive	12	3	18	18	0
Notebook, monopoly	8	4	8	24	4
Swift, competitive	15	150	1.125	1.125	0
Swift, monopoly	10	200	0.5	1.5	0.25

- Example (F25): Swift's average price = \$2 billion $\div$ 10 million tickets = \$200. With no fixed cost, total cost = area under $MC = \frac{1}{2} \times 10 \text{ million} \times \$100 = \$0.5 \text{ billion}$, so profit = \$2 billion – \$0.5 billion = \$1.5 billion. The key also accepted \$1 billion.
- A monopoly has no supply curve. If demand rises, P and Q rise. If MC rises, Q falls and P rises. A fixed cost changes only profit. If profit turns negative, the firm exits.
- Patents give a temporary monopoly so its profit pays back costly research. When one ends (the drug Humira, 2023), rivals enter and P falls toward MC.
- Policy: antitrust law, price regulation of utilities, government ownership (USPS), and limits on patents. The 2024 Google case (Problem Set 2) was about dominance and ad prices, not consumer prices.

Monopoly: Q from $MR = MC$, then P from demand



Monopoly: $MR = MC$, P from D

Negative externality

Externalities

- An externality is a cost or benefit that falls on people outside the market. The market price leaves it out.
- $SMC = PMC + EMC$ and $SMB = PMB + EMB$ (S = social, P = private, E = external).**
- The market produces where $PMB = PMC$. The efficient quantity Q^* is where $SMB = SMC$.

	Negative	Positive
Gap	SMC above PMC	SMB above PMB
Market Q	too much	too little
Examples	gas, smoking, noise	vaccines, schooling
Pigouvian fix	tax = EMC at Q^*	subsidy = EMB at Q^*

- DWL is the triangle between SMC and SMB from the market Q to Q^* , pointing at Q^* . If EMC is constant, $DWL = \frac{1}{2} \times EMC \times (\text{market } Q - Q^*)$.
- External cost at any Q is the area between SMC and PMC ($EMC \times Q$ if EMC is constant). Total surplus = CS + PS – external cost.
- A Pigouvian tax equal to EMC lifts supply up to SMC, so the market lands on Q^* and the DWL disappears. Tax revenue = tax $\times$ Q^* . Problem Set 2 tests this: a \$2 external cost needs a \$2 tax.
- TRAP** Mistake: optimal pollution is zero. Correct: optimal Q is where $SMC = SMB$, usually above zero (F25).
- Coase: with clear property rights and cheap bargaining, the people affected can fix an externality privately.
- Ostrom: groups that deal with each other for years can fix it with trust and their own rules.
- Cap and trade: the government issues a fixed number of tradable permits. Firms that cut pollution cheaply sell theirs, so cuts happen where they cost least.
- The optimal carbon tax equals marginal damage. Most damage comes later, so heavy discounting of the future makes it small.

Problem Set 2, Q3: demand $P = 5 - Q$, flat $MC = 1$

Case	Q	P	CS	PS	DWL	Notes
Competitive	4	1	8	0	0	Demand $\epsilon = -0.25$ (inelastic).
\$2 tax	2	3	2	0	2	Buyers pay all of it (flat supply). Revenue 4.
\$2 pollution	4	1	8	0	2	$SMC = 3$, so $Q^* = 2$. External cost 8. A \$2 tax fixes it.
Monopoly	2	3	2	4	2	$MR = 5 - 2Q$. PS = profit = $(3 - 1) \times 2 = 4$ (the key wrongly says 0).
Monopoly, pollution	2	3	2	4	0	Efficient by luck: $SMC = 3$ meets demand at $Q = 2$. External cost 4.

F25 midterm Q2: eggs

2a-d	Demand is steep because nothing replaces eggs 1 for 1. MC (feed, water, hen upkeep) is low and rises slowly, so supply is low and flat. Costly California land is a big fixed cost: a big barrier to entry over 1 to 2 years, not as big over 5.
2e-g	The curves cross at \$2 and 10 billion dozen. Prices spiked in January 2023 and January 2025 but not 2024, after fall bird migration spread flu. That is a supply shock, since a baking boom would not skip a year. Supply shifts left (up), so P jumps to about \$6 and Q falls only a little (to about 8 billion).
2h	Trader Joe's sells mostly non-egg goods, so the shock barely moved its costs. It raised its egg price but kept it below market, with a one-dozen limit. Eggs are in most Denny's dishes, so its MC jumped and it added a surcharge.